



Adaptive Inference: Orchestrating Fine-Tuned LLMs with Serverless GPUs in HPC Environments

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RESEARCH PROBLEM

Fine-tuning and hosting LLMs is becoming feasible for many organizations, yet:

- Most still rely on static deployments or external APIs
- Self-hosting multiple LLMs is expensive to run 24/7
- Failures or poor outputs lack fallback mechanisms
- Data privacy is at risk with public endpoints
- Infrastructure is frequently overprovisioned, wasting compute resources
- Reproducibility is difficult when using black-box APIs outside your control

ARCHITECTURE

